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Review

Efficacy of triamcinolone acetonide combined with botulinum toxin A in the treatment of hypertrophic scars and keloids: A meta-analysis

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ABSTRACT

Background: This meta-analysis aims to evaluate the efficacy and safety of triamcinolone acetonide (TCA) combined with botulinum toxin type A (BTA) for treating hypertrophic scars and keloids.

Methods: A comprehensive search of randomized controlled trials published before September 2023 was conducted across the Cochrane Library, Embase, PubMed, Web of Science, and CNKI databases. The analysis involved calculating pooled weighted mean difference (WMD), pooled risk ratios (RR), and 95 % confidence intervals (CI).

Results: Inclusive of 11 studies with a total of 561 patients, the meta-analysis revealed a statistically significant difference in the effective rate between the BTA+ TCA and control groups (RR = 1.28, 95 % CI = 1.14–1.44). Moreover, BTA+ TCA demonstrated a significant improvement in Visual Analog Scale scores (WMD = −1.69, 95 % CI = −2.72 – −0.66) and Vancouver Scar Scale scores (WMD = −1.46, 95 % CI = −1.90 – −1.02) compared to the control group. However, no statistically significant difference in scar thickness was observed between the BTA+ TCA and control groups (WMD = −0.11, 95 % CI = −0.30 - 0.09).

Conclusion: This meta-analysis showed that the combined use of BTA and TCA demonstrates high effectiveness in scar treatment, but its influence on scar thickness is limited. Future research should further explore the sources of heterogeneity and validate the long-term effects and safety of this therapy.

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Contents

1. Introduction

Hypertrophic scars and keloids are benign fibroproliferative disorders that typically arise in the dermal layer after skin injury or irritation [1]. Both types of scars present as firm, raised lesions that may be asymptomatic or associated with itching, pain, and even restricted movement, yet they exhibit distinct differences [2]. Hypertrophic scars generally remain within the confines of the original wound and may gradually regress over time [3]. In contrast, keloids are more severe, do not regress with time, and may extend beyond the original injury site, with a tendency to recur [4]. Hypertrophic scars commonly occur in areas of high tension, while keloids predominantly appear on earlobes, shoulders, sternum, and upper back and have a familial predisposition [5]. These scar tissues not only affect aesthetics but also cause discomfort such as itching, pain, and swelling, leading to psychological barriers and a diminished quality of life for patients [6,7]. Various invasive and non-invasive treatment modalities have been reported for managing these scars. Conservative therapy is the first-line treatment for hypertrophic scars, including topical or intralesional injection with corticosteroids, pressure therapy, and silicone gel sheets [8,9]. For keloids of small to medium size (< 20 cm^2), most cases also employ conservative treatments, such as intralesional corticosteroid injections, corticosteroid tape/plasters, or intralesional chemotherapy drugs [10]. However, a standardized treatment protocol has yet to be established in clinical practice.

Triamcinolone acetonide (TCA), a corticosteroid, has been widely used for the treatment of hypertrophic scars. It acts by reducing inflammation, inhibiting collagen synthesis, and promoting collagen degradation [11]. Botulinum toxin type A (BTA), a neurotoxin, functions by inhibiting impulse transmission at the neuromuscular junction [12]. Relevant studies have shown that BTA not only suppresses local muscle activity and reduces skin tension but also inhibits fibroblast proliferation, thus curbing scar growth [13,14]. While recent randomized controlled trials (RCTs) have provided some evidence suggesting the superiority of a combination of TCA and BTA compared to TCA alone, these studies have generally involved small sample sizes, and no definitive evidence has emerged to support a conclusive analysis [15–17]. To shed light on whether a relative benefit of combining TCA with BTA exists in the treatment of hypertrophic scars and keloids, we conducted a systematic review and meta-analysis of all published RCTs evaluating TCA in combination with BTA for the treatment of these scars.

2. Methods

This meta-analysis was performed based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Table S1), and this meta-analysis was registered on PROSPERO (CRD42024546670).

2.1. Search strategy

A comprehensive search of the Cochrane Library, Embase, PubMed, Web of Science, and CNKI databases was conducted for the therapeutic effects of BTA+ TCA on keloids and hypertrophic scars up until June 2024. The search was performed using specific text words based on medical terms, with the keywords used were "Keloid," "Hypertrophic Scar," "Scar," "Triamcinolone Acetonide," and "Botulinum Toxin Type A." Additionally, to ensure comprehensive coverage, the reference lists of identified RCTs were manually examined for potentially relevant papers. The detailed search strategy is shown in Table S2.

2.2. *Eligibility criteria*

The inclusion criteria for this study were as follows: (1) RCTs or controlled trials (CTs) without any language limitations; (2) patients with hypertrophic scars or keloids; (3) comparison of TCA combined with BTA versus TCA or BTA alone in the treatment of keloid and hypertrophic scars; and (4) inclusion of any of the following outcome indicators: effective rate (effective patient number divided by the total patient number), visual analog scale (VAS), Vancouver Scar Scale (VSS), and scar thickness. The exclusion criteria were: (1) non-randomized controlled studies, case reports, conference abstracts, and animal or cell biology research; (2) use of other methods or drugs for the treatment of hypertrophic scars or keloids; inability to extract outcome data; and (3) absence of relevant outcome data.

2.3. *Data extraction and quality assessment*

Two independent reviewers (SJZ and XJR) extracted data using a predefined data extraction form. The extracted data included the first author, year of publication, participant characteristics (such as mean age, proportion of male participants, and sample size), intervention and control groups, and measured outcomes. In cases of disagreement, a third reviewer (CYM) was consulted for analysis and resolution.

The risk of bias was evaluated according to the Cochrane Handbook and analyzed using Review Manager 5.3. The parameters assessed included random sequence generation, allocation concealment, blinding, incomplete outcome data, and selective reporting. *Statistical Analysis*.

All analyses were conducted using STATA version 15.0 (StataCorp). For dichotomous variables, the inverse variance risk ratio (RR) and corresponding 95 % confidence intervals (CI) were calculated to evaluate the effect of the intervention. For continuous data comparisons, the weighted mean difference (WMD) and 95 % CI were calculated. Heterogeneity between studies was assessed using the I^2 statistics. The random-effects model was applied when $I^2 > 50$ % indicating substantial heterogeneity, while the fixed-effects model was used when $I^2 < 50$ % indicating low heterogeneity. Meta-regression analysis was performed to evaluate the potential source of heterogeneity among studies. Additionally, Egger's test was applied to assess publication bias and exclude its potential impact on the results.

3. Results

3.1. *Search results and study characteristics*

The initial search yielded 101 studies, of which 11 was excluded as duplicates, leaving 90 studies. After screening the titles and abstracts, 67 studies were removed based on the inclusion and exclusion criteria. After reading the full text of the remaining 23 studies, 12 studies were further excluded. Finally, 11 studies [15–25] involving 561 patients were identified and included in the meta-analysis (Fig. 1). All the studies were published between 2014 and 2023. The sample size in each study ranged from 18 to 88. The basic characteristics of all the studies are presented in Table 1. The studies included in this meta-analysis were of moderate quality, with most having a moderate to low risk of bias (Fig. 2).

3.2. *Meta-analysis results*

3.2.1. *Effective rate*

Six studies reported effective Rate. A heterogeneity test showed no heterogeneity among the studies ($I^2 = 0$ %, $P = 0.988$); therefore, the fixed-effect model was used. The effective rate of the BTA+ TCA group was significantly higher than that of the control group (RR = 1.28, 95 % CI = 1.14–1.44) (Fig. 3).

3.3. *VAS score*

The VAS is frequently utilized to assess pain and itching levels, as indicated by VAS scores ranging from 0 (representing no symptoms) to 10 (indicating extremely severe symptoms). Six studies provided data on VAS scores. A significant level of heterogeneity was observed ($I^2 = 95.7$ %, $P < 0.0001$), necessitating the use of a random-effect model. The results indicated that the VAS score for the BTA+ TCA group was notably lower than that for the control group (WMD = −1.69, 95 % CI = −2.72 – −0.66) (Fig. 4).

3.4. *VSS score*

VSS score is derived by summing the individual scores of various indicators, including vascularity, pigmentation, pliability, and height, to evaluate the severity and characteristics of scars. A lower VSS score indicates that the scar more closely resembles normal skin, whereas a higher score suggests greater severity and deviation from normal skin. Five studies have reported VSS scores, but a meta-analysis revealed significant heterogeneity among these studies ($I^2 = 68.8$ %, $P = 0.012$). Consequently, a random-effects model was employed. The analysis showed that the combination of BTA and TCA significantly improved VSS scores compared to the control group (WMD = −1.46, 95 % CI = −1.90 to −1.02) (Fig. 5).

3.5. *Scar thickness*

Scar thickness data were derived from two studies. With low heterogeneity ($I^2 = 49.9$ %, $P = 0.158$), a fixed-effect model was employed. The analysis revealed no statistically significant

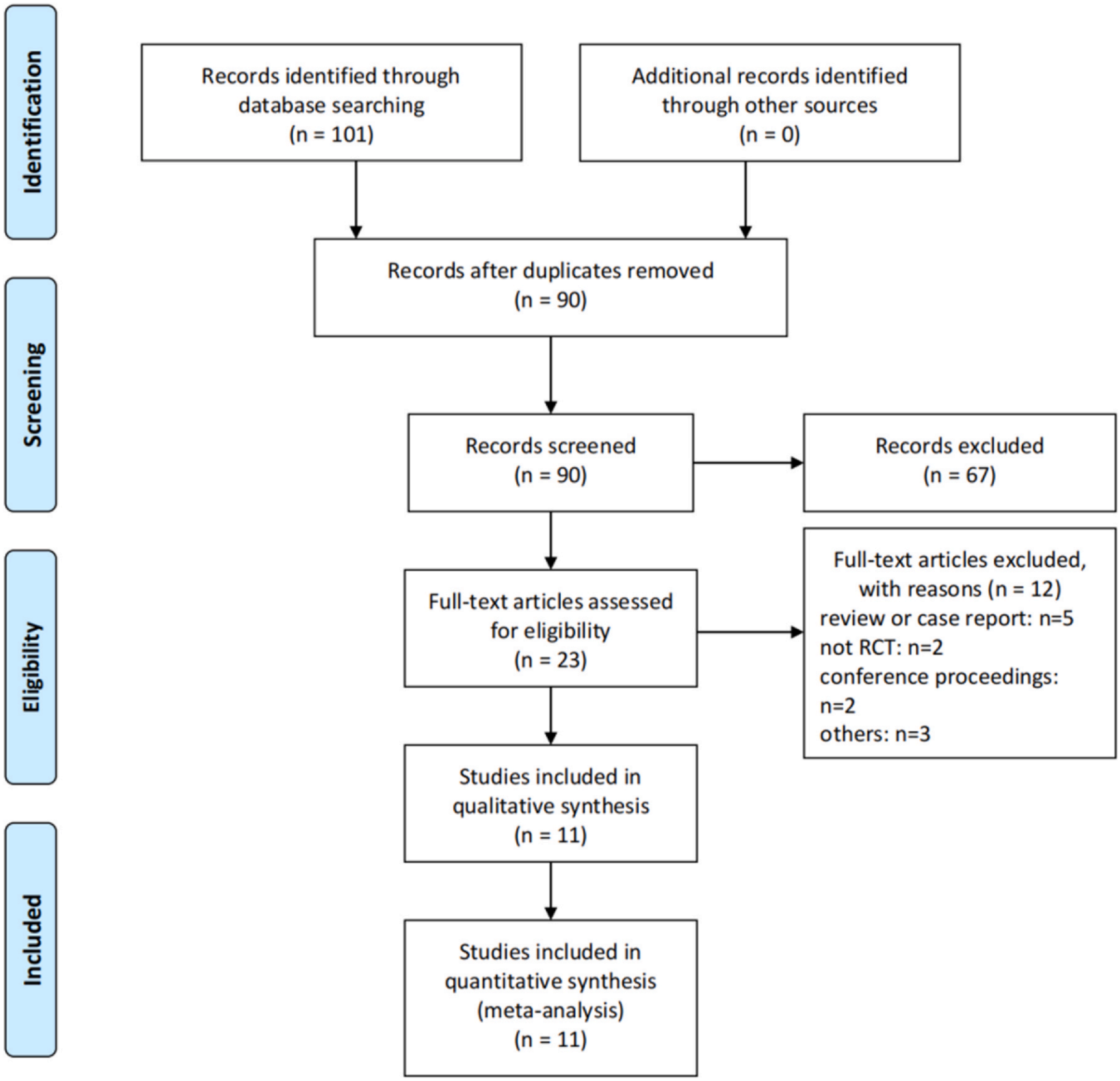

Fig. 1 – Flow chart showing the selection process of the studies in the meta-analysis.

enhancement in scar thickness between the groups receiving BTA+ TCA and those receiving TCA or BTA alone (WMD = −0.11, 95 % CI = −0.30 – 0.09) (Fig. 6).

3.6. *Meta-regression*

A meta-regression was performed to pinpoint the source of heterogeneity in VAS and VSS, utilizing sample size, year of publication, and follow-up time as covariates. For VAS, the analysis indicated that sample size ($P = 0.37$) did not significantly contribute to the heterogeneity among studies. However, publication year and follow-up duration were identified as potential sources of the observed heterogeneity ($P < 0.05$, Table 2). Regarding VSS, the regression analysis results showed that sample size, publication year, and follow-up duration could all be potential sources of the observed heterogeneity ($P < 0.05$, Table 2).

3.7. *Publication bias and sensitivity analysis*

The Egger tests ($P > 0.05$) revealed no significant evidence of publication bias. Furthermore, a sensitivity analysis was conducted by systematically removing studies to assess the impact of individual studies on the overall effect estimate. The findings demonstrated that the exclusion of any single study did not exert a noteworthy influence on the combined effects.

4. Discussion

Keloids and hypertrophic scars are often considered as abnormal fibroblast proliferation during the wound healing process, with a genetic predisposition [26]. Hypertrophic scars and keloids are two different types of scars, each with

Table 1 – Basic characteristic of included RCT studies in the Meta-analysis.

Study	Country	Age (years)	Sex (male/ female)	Treatment (experimental group)	Treatment (control group)	BTA Brand	Source of Scars	Lesions Treated	Follow-up	Outcomes measurement
Li 2014	China	27 ± 7.9	21/17	TCA (5 mg/cm, maximum dose not exceeding 80 mg)+BTA(25 IU/mL, maximum dose not exceeding 100 iu once) was injected once every two weeks, and five times was a course of treatment	The dosage of TCA injection is 5 mg per square centimeter, and the maximum dosage is not more than 80 mg, once every two weeks, and five times is a course of treatment.	N/A	N/A	Hypertrophic scar	6 m	VAS
Cheng 2015	China	27.5 ± 4.3	21/25	BTA (1 U/ point, maximum dose not exceeding 50 mg once) was injected at multiple points with an interval of 1 cm, and TCA was injected with a high-pressure syringe 2 mg /cm two weeks later, with no more than 50 mg at a time	TCA was injected at a dose of 5 mg /cm^2, with the maximum dose not exceeding 80 mg, once every 2 weeks, 5 times per course of treatment	N/A	Early proliferative scars due to different causes	Hypertrophic scar	6 m	Effective rate, VAS
Li 2016	China	18 –35	33/41	BTA (1 U/ point, maximum dose not exceeding 50 mg once) was injected at multiple points with an interval of 1 cm, and TCA was injected with a high-pressure syringe 2 mg /cm two weeks later, with no more than 50 mg at a time	TCA was injected at a dose of 5 mg /cm^2, with the maximum dose not exceeding 80 mg, once every 2 weeks, 5 times per course of treatment	N/A	N/A	Hypertrophic scar	6 m	Effective rate, VAS
Wang 2017	China		45/43	BTA (1 U/ point) was injected at multiple points with an interval of 1 cm, and TCA was injected with a high-pressure syringe 2 mg /cm two weeks later, with no more than 50 mg at a time	TCA was injected at a dose of 5 mg /cm^2, with the maximum dose not exceeding 80 mg, once every 2 weeks, 5 times per course of treatment	N/A	N/A	Hypertrophic scar	6 m	Effective rate, VAS

(continued on next page)

Table 1 – (*continued*)

Study	Country	Age (years)	Sex (male/ female)	Treatment (experimental group)	Treatment (control group)	BTA Brand	Source of Scars	Lesions Treated	Follow-up	Outcomes measurement
Rasaii 2019	Iran	23.3 ± 1.2	11/12	TCA 20 mg/mL + BTA 20 u/ mL, once every 4 weeks, 3 times in total	TCA 40 mg/mL, once every 4 weeks, 3 times in total	Neuronox	N/A	keloid	11.2 m	VSS
Liu 2020	China	18 –54	20/10	Thirty min after TCA 40 mg injection, BTA (2.5 U/ point) was injected at 1 cm intervals, with a total dose of less than 20.0 U/ time, once every 4 weeks, 5 times in total	TCA 40 mg, single dose < 80 mg, once every 4 weeks, 5 times in total	Lanzhou Institute of Biological Products Co., Ltd.	N/A	keloid	26.97 m	Effective rate, VAS, VSS
Gamil 2020	Egypt	19 –43	34/16	BTA injections were done 0.1 mL until clinically visible slight blanching, the dosage was adjusted to 2.5 IU cm^3 of the lesion with a maximum dosage of 100 I session, followed by the TCA injections were done at a dose of 0.1- cm^3 of involved skin with a maximum dosage of the session	The TCA injections were done at a dose of 0.1- cm^3 of involved skin with a maximum dosage of the session	N/A	N/A	keloid	6 m	Thickness
Che 2021	China	12 –47	35/23	BTA (5 U/ point) was injected at multiple points with an interval of 2 cm, and TCA was injected with 5 mL four weeks later, with no more than 50 mg at a time	TCA was injected at a dose of 5 mL, with the maximum dose not exceeding 50 mg, once every 1 weeks, 4 times per course of treatment	Heng Li	Burns, surgery, trauma, and other causes	Hypertrophic scar	1 m	VAS, VSS
Huang 2021	Taiwan	35.72 ± 14.19	7/11	The TCA/BTA regimen was prepared with BTA at 4 U in 0.1 mL mixed with the same TCA 0.1 mL and 2 % lidocaine HCl 0.1 mL, intralesional injection was administered over a total of 3 sessions, at 4-week intervals	The TCA regimen was prepared with 0.1 mL of TCA, mixed with 0.1 mL 2 % lidocaine HCl and an additional 0.1 mL normal saline, intralesional injection was administered over a total of 3 sessions, at 4-week intervals	Botox	Surgery or trauma	hypertrophic scars and keloids	3 m	VAS, VSS, Thickness

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Table 1 – (continued)

Study	Country	Age (years)	Sex (male/ female)	Treatment (experimental group)	Treatment (control group)	BTA Brand	Source of Scars	Lesions Treated	Follow-up	Outcomes measurement
Yu 2022	China	13 –49	35/41	The dosage of TCA single injection should not exceed 80 mg. BTA (≤5 U/ point) should be injected 1 h after TCA injection, once every two weeks, and five times is a course of treatment.	BTA (≤5 U/ point) was injected, with a single injection dose not exceeding 100 U, once every two weeks, and five times is a course of treatment.	Lanzhou Bio-Technology Development Co., Ltd.	Burns	Hypertrophic scar	1 m	Effective rate, VAS, VSS
Liu 2023	China	16 –70	27/33	BTA ≤ 100 U/ time, TCA ≤ 40 mg/ time, once every four weeks, for a total of three times	TCA was injected once every 4 weeks with a single dose of less than 40 mg, and the treatment was conducted for 3 times	N/A	N/A	keloid	3 m	Effective rate

VAS: visual analog scale, VSS: vancouver Scar Scale, BTA: Botulinum toxin type A, TCA: Triamcinolone acetonide, N/A: Not Applicable

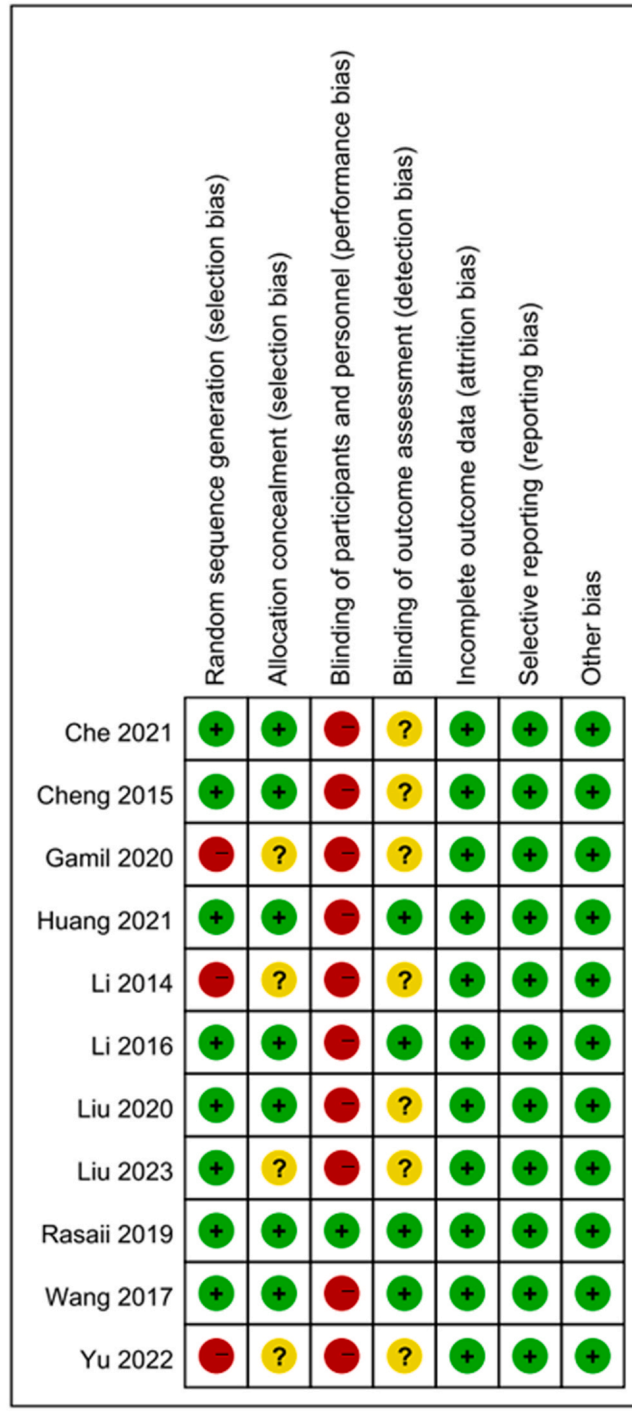

Fig. 2 – Risk of bias summary. Green means 'low risk', red means 'high risk' and yellow means 'unclear risk'.

distinct formation and regression patterns. Hypertrophic scars are formed due to excessive collagen deposition during the wound healing process and are usually confined to the boundaries of the original wound, not extending into normal skin [27]. These scars may naturally regress, becoming flatter and softer, and their color may gradually blend with the surrounding skin within 6 to 18 months after wound healing [6,7]. Treatment methods include pressure therapy, laser

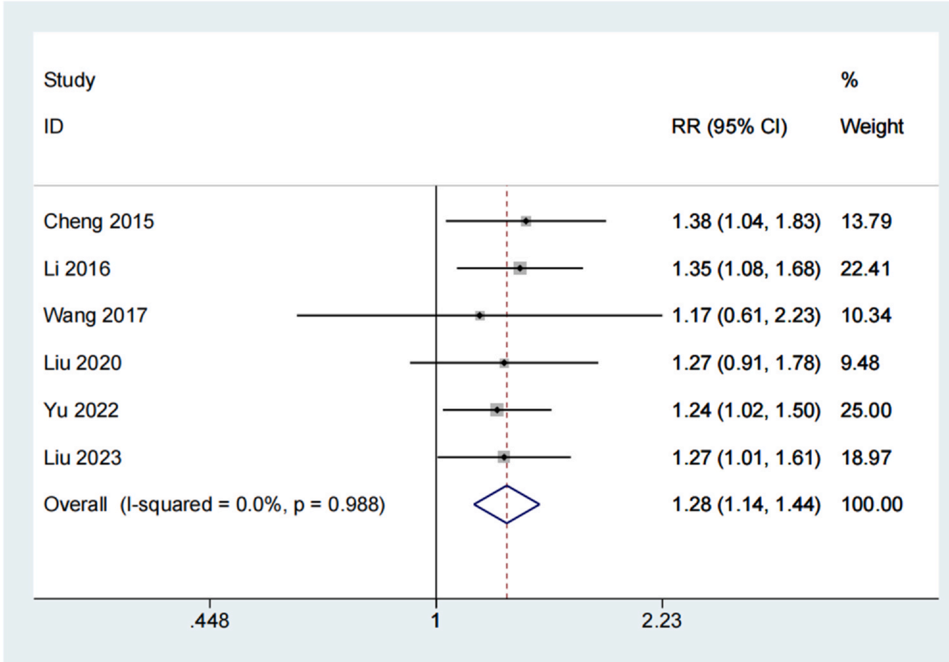

Fig. 3 – Forest plot of effective rate between Triamcinolone acetonide (TCA) + Botulinum toxin type A (BTA) and TCA groups.

therapy, corticosteroid injections, and surgical excision. In contrast, keloids are raised, hard scars formed due to excessive collagen production that extends beyond the boundaries of the original wound, often accompanied by itching or pain [28,29]. Keloids typically do not regress naturally and may become more persistent and prominent over time. Treatment methods include corticosteroid injections, laser therapy, surgical excision, radiation therapy, silicone sheets, and pressure therapy [30–33]. TCA is a potent corticosteroid that prevents collagen synthesis and accelerates the breakdown of collagen fibers [11]. It is commonly used to treat scar hyperplasia and prevent scar recurrence. Although TCA has

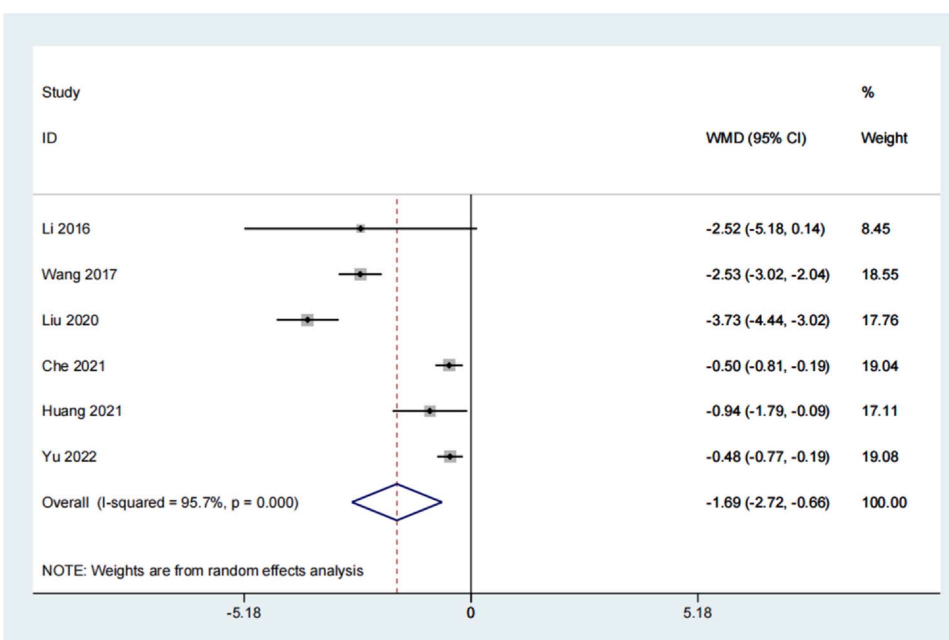

Fig. 4 – Forest plot of visual analog scale score between TCA + BTA and TCA groups.

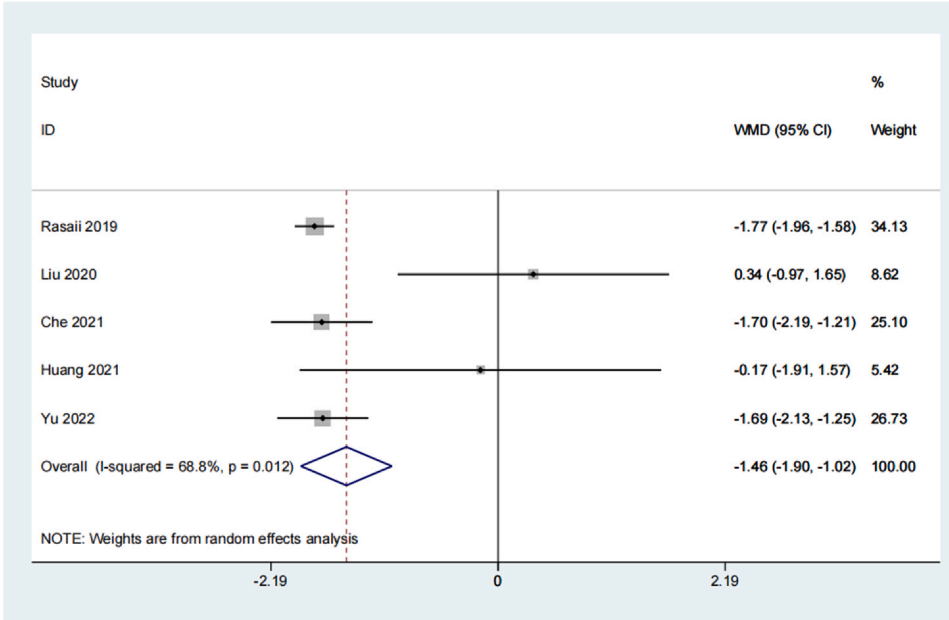

Fig. 5 – Forest plot of vancouver Scar Scale score between TCA + BTA and TCA groups.

demonstrated good clinical efficacy in the treatment of hypertrophic scars, it has been observed that the recurrence rate is high when TCA is used alone, and the treatment response for symptoms such as scar pain and itching is low in some patients [34,35]. With the advancement of clinical research, the combination of medications has become a new approach for scar treatment. BTA is a neurotoxic drug that acts on cholinergic nerve terminals, interfering with acetylcholine release by antagonizing calcium ion action, inhibiting muscle fiber contraction, and causing muscle relaxation and paralysis[36]. Recent studies have found that BTA has an inhibitory effect on scar hyperplasia and can effectively alleviate pain and itching symptoms in patients with early-stage hypertrophic scars, leading to scar regression and softening, resulting in improved cosmetic outcomes of surgeries [37,38]. Previous studies have reported favorable

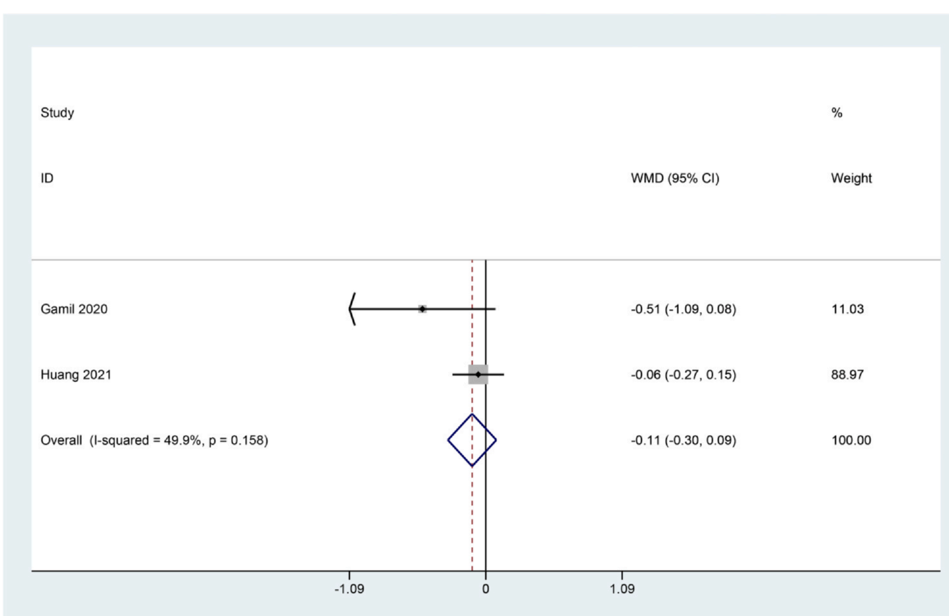

Fig. 6 – Forest plot of scar thickness between TCA + BTA and TCA groups.

Table 2 – Meta-regression of factors affecting between-study heterogeneity.

Heterogeneity	Coefficient	Std. Err.	z	P > z	[95 % Conf. Interval]
VAS					
Sample size	−0.0055203	0.0061574	−0.9	0.37	−0.0175886 to 0.006548
publication year	0.2827493	0.0664035	4.26	0.000	0.1526008 to 0.4128979
Follow up	−0.1165969	0.0184452	−6.32	0.000	−0.1527489 to −0.0804449
VSS					
Sample size	−0.0420226	0.0211061	−1.99	0.046	−0.0833897 to −0.0006555
publication year	1.068214	0.3716244	2.87	0.004	0.339844 to 1.796585
Follow up	0.0731665	0.0327261	2.24	0.025	0.0090244 to 0.1373085

results with the combination of TCA and BTA, and this combination has been used in clinical practice [21–23]. This meta-analysis, based on eleven studies, demonstrated that the combined use of TCA and BTA significantly improved scar recovery, indicating that the combination of TCA and BTA is more effective in scar treatment compared to TCA alone.

The combination of TCA and BTA may act on scarring through multiple mechanisms. TCA is a widely used corticosteroid for scar treatment, which inhibits the proliferation of scar fibroblasts, enhances collagen degradation, reduces scar tissue atrophy, and decreases cell migration [11]. However, the use of TCA also has potential side effects such as skin atrophy, pigmentation changes, folliculitis, and skin infections [39]. Additionally, for some refractory scars, the treatment efficacy of TCA may be unsatisfactory [40]. On the other hand, BTA has a significant impact on fibroblast activity, primarily by inhibiting fibroblast proliferation, reducing collagen deposition, and suppressing the synthesis and secretion of transforming growth factor-β (TGF-β) [41]. However, there was no significant difference in scar thickness between the two groups, which may be due to the complexity of scar formation. Scar formation involves multiple cell types and signaling pathways, and although BTA has a notable effect on fibroblasts[42], the influence of other cells and factors may compensate in the formation of scar thickness. Additionally, scar formation and maturation is a long-term process, and the observation period of the study may not be sufficient to reflect the inhibitory effects of BTA. Even if initial collagen deposition is reduced, the later remodeling process might result in no significant change in scar thickness. Measurement limitations and individual differences could also contribute to the lack of significant change in scar thickness. Moreover, the dosage and administration method of BTA might affect its efficacy. In summary, although BTA has a clear inhibitory effect on fibroblast activity, the lack of significant difference in scar thickness may be the result of multiple factors acting together. Further research and longer observation periods may help to more comprehensively understand the impact of BTA on scar formation. When TCA and BTA are used in combination, their effects may be mutually enhanced. For example, in a rat model, the TCA +BTA treatment regimen reduced the expression of protein gene product (PGP) 9.5, nerve growth factor (NGF), phosphorylated signal transducer and activator of transcription 3 (p-STAT3), and transient receptor potential vanilloid 1 (TRPV1), restored the level of kappa-opioid receptor, and alleviated post-burn pruritus [43]. These results suggest that BTA may serve as a potential adjunct to TCA by inhibiting the release of neuropeptides and inflammation associated with neurogenic itching, thus enhancing and maintaining the effect of TCA. Similarly, the therapeutic effect of BTA combined with TCA is superior to using BTA alone. Research results indicate that the combined use of BTA and TCA shows higher efficacy in the treatment of hypertrophic burn scars. These two drugs improve the severity of scars by influencing the dynamic balance of fibroblasts and collagen. Furthermore, Huang et al. found that the skin neural system also promotes the formation of hypertrophic scar tissue by increasing nerve density and upregulating NGF expression [23]. The combined treatment of TCA and BTA can strongly inhibit NGF release, limit nerve fiber growth, and reduce TRPV1 expression. Therefore, the reduction in itching sensation is accompanied by a decrease in scar thickness. Therefore, although TCA has some limitations in scar treatment, combining it with BTA may help address these issues and improve treatment efficacy. However, further research is needed to validate and optimize this combination therapy.

The combined use of botulinum toxin and triamcinolone is influenced by the follow-up period. Short-term follow-ups (1–6 months) primarily focus on immediate effects such as reduced scar thickness, pain, and itching relief, but may not reflect long-term effects and recurrence. Medium to long-term follow-ups (11.2–27 months) can assess the lasting effects of the treatment and the stability of the scars, as shown in studies by Rasaii (2019) [17] and Liu (2020) [21], which indicate good long-term outcomes and reduced scar recurrence. Therefore, it is crucial to consider the differences in follow-up periods comprehensively to avoid misleading conclusions. The age of the scar may also affect the treatment outcome, with newer scars potentially responding more significantly and older scars showing slower improvement. The type, size, location of the scar, and individual differences may also influence the treatment effect. The combined use of botulinum toxin and TCA may provide comprehensive effects, but the relationship between efficacy and the age of the scar requires further research and long-term follow-up to be determined.

Our study indicates that the combined treatment of BTA and TCA was significantly more effective in alleviating pain and itching compared to the control group, as demonstrated by the VAS scores. However, despite the statistical significance, substantial heterogeneity was observed among the six studies. This high degree of heterogeneity suggests significant differences in study design, participant characteristics, or interventions. Regression analysis indicated that the sample size did not significantly contribute to the heterogeneity among studies. However, publication year and follow-up duration were identified as potential sources of the observed heterogeneity. Therefore, these results should be interpreted with caution. Future research should investigate the sources of heterogeneity and aim to reduce variability through standardized designs, enhancing the reliability and reproducibility of the findings. The VSS is an effective tool for evaluating the severity and characteristics of scars, with lower VSS scores indicating that the scar is closer to normal skin. Despite significant heterogeneity among the five studies on VSS scores, the analysis results showed that the combined use of BTA and TCA significantly improved VSS scores, indicating that this treatment method effectively improved scars, making them closer to normal skin. Notably, the regression analysis results concerning VSS scores indicated that sample size, publication year, and follow-up duration could all be potential sources of the observed heterogeneity. This provides valuable insights for the clinical treatment of scars and may improve the condition of patients' scars in practical applications. The Egger tests revealed no significant evidence of publication bias. Furthermore, a sensitivity analysis was conducted by systematically removing studies to assess the impact of individual studies on the overall effect estimate. The findings demonstrated that the exclusion of any single study did not exert a noteworthy influence on the combined effects.

The economic impact of using BTA for scar treatment requires a comprehensive consideration of direct costs, indirect costs, long-term benefits, patient satisfaction, insurance policies, cost-effectiveness analysis, and market size. The medication cost of BTA is relatively high and requires regular injections, increasing long-term expenses. Additionally, BTA treatment necessitates the involvement of specialized medical personnel, which raises the operational costs of clinics. However, significant effectiveness of BTA in reducing scar hypertrophy and improving scar appearance may reduce the need for subsequent treatments, lower recurrence rates, and enhance patient satisfaction, thereby proving cost-effective in the long term [44]. In contrast, traditional treatments such as laser therapy, surgical excision, and topical medication have lower initial costs but may require repeated treatments, increasing long-term expenses. Insurance coverage also affects the economic feasibility of BTA treatment, with some patients potentially facing higher out-of-pocket costs. Nevertheless, with the growing demand for cosmetic and medical treatments and increased market competition, the cost of BTA treatment may decrease, improving its accessibility. Overall, despite the higher initial cost of BTA treatment, its significant efficacy and potential long-term economic value make it a reasonable option in certain cases. Detailed cost-effectiveness analyses are needed to evaluate its economic rationality in different contexts.

However, this study also has some limitations that should be carefully considered when interpreting the results. Firstly, the eleven included articles in this study mainly employed randomized controlled study designs; however, the detailed description of blinding and allocation concealment for the randomization process was lacking, which may result in a high risk of selection bias. Secondly, in randomized controlled trials comparing injection treatments, it can be challenging to recruit a sufficient number of patients. Sometimes, patients may prefer faster treatment methods, such as surgical excision. Therefore, the number of studies included and the sample size are both small. Thirdly, most of the included studies were conducted and reported in Asia, with the majority of patients being Asians. Therefore, the conclusions of this meta-analysis may not represent patients of other ethnicities. Finally, most of the included studies did not report data on the subcomponents of the Vancouver Scar Scale (VSS), precluding the possibility of conducting a detailed subgroup analysis for each component. Therefore, additional high-quality clinical trials are warranted to further evaluate the efficacy of BTA+TCA.

In conclusion, the results showed that the BTA+TCA group significantly outperformed the control group in terms of efficacy, VAS scores, and VSS scores, indicating that this combined treatment approach has notable advantages in improving scar treatment outcomes, reducing patient pain and itching, and enhancing the appearance and texture of scars. However, the impact on scar thickness was not significant. Overall, the combined use of botulinum toxin and trichloroacetic acid demonstrates high effectiveness in scar treatment, but its influence on scar thickness is limited. Future research should further explore the sources of heterogeneity and validate the long-term effects and safety of these results.

Ethics approval

Not applicable.

Consent to participate/publish

Not applicable.

Code availability

Not applicable.

Funding

None.

CRediT authorship contribution statement

XJR and CYM contributed to the study conception and design. Material preparation, data collection, and analysis were performed by SJZ, ZSQ and ZZY. The first draft of the manuscript was written by SJZ. The manuscript was revised by SSY and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data Availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declaration of Competing Interest

The authors declare that they have no conflicts of interest.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.burns.2024.08.016.

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